THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 04

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** 0	every point on the x-axis has $y = 0$
2	**(d)** $5n + 2$	common difference 5, so $a_n = 5n + 2$
3	**(c)** $\frac{2}{3}$	standard method $\Rightarrow \frac{2}{3}$
4	**(d)** $25x^2 + 20x + 4$	$(a+b)^2 = a^2 + 2ab + b^2$
5	**(c)** $(x - 7)(x + 6)$	find a, b with $a + b$ = middle coefficient and ab = constant
6	**(a)** 50°	angle at the centre is double the angle on the circle
7	**(d)** 22 cm	$2\pi r = 2 \times \frac{22}{7} \times 3.5 = 22$
8	**(d)** 600	expected number $= P \times$ population
9	**(a)** 210	$\frac{20 \cdot 21}{2}$
10	**(c)** 3 factors	p^2 has $1, p, p^2$
11	**(d)** A is false but R is true	A is false; R is a true fact that in fact contradicts A
12	**(a)** Both A and R are true and R is the correct explanation of A	equal weights
13	**(c)** 82	$\frac{82 \times 1 + 74 \times 2 + 90 \times 2}{5} = 82$
14	**(c)** 12	the rightmost element is the gcd
15	**(c)** Āryabhaṭa	Āryabhaṭīya, 499 CE
16	**(c)** 360°	full turn around a vertex
17	**(c)** rises as we move from left to right	rise and run have the same sign
18	**(d)** (2, 5)	substitute the pair in both equations
19	**(b)** 54 cm^2	side $= 3$ cm, TSA $= 6a^2 = 54$
20	**(c)** one-third of	salt-filling experiment: three cones fill one cylinder

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	$AB = \sqrt{4+4} = 2\sqrt{2}$, $BC = 2\sqrt{2}$, $AC = \sqrt{16+16} = 4\sqrt{2}$. Since $AB + BC = AC$, the points are collinear.	1+1
22	$x + (x + 10) = 64 \Rightarrow x = 27$; numbers 27 and 37.	1+1
23	$\angle A + \angle C = 180^\circ$, so by the converse (Theorem 12) the vertices are concyclic.	1+1
24	(i) $\frac{4}{52} = \frac{1}{13}$. (ii) $\frac{2}{52} = \frac{1}{26}$.	1+1
25	Proposition: If a person lives in Jaipur, then the person lives in Rajasthan (1). Converse: If a person lives in Rajasthan, then the person lives in Jaipur — false (a person in Jodhpur) (1).	1+1
26	Stacked: bar lengths show actual values and totals differ (1). 100% stacked: every bar has the same length and shows only the shares (1).	1+1
27	The tiles meeting at a point fit around it with no gap or overlap, i.e. they complete a full turn (2).	2
28	Put $y = 0$: $x = 4$, so $(4, 0)$ on the x-axis (1). Put $x = 0$: $y = 3$, so $(0, 3)$ on the y-axis (1).	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	(i) $\frac{2}{3} \times 36 = 24$ m (1). (ii) $\frac{1}{3} \times 36 = 12$ m (1). (iii) The Centroid Theorem — medians are concurrent and divide in the ratio $2 : 1$ (1).	1+1+1
30	Area $= 4\pi r^2 = 4 \times \frac{22}{7} \times 49$ (1) $= 616$ cm^2 (1). Cost $= 616 \times 0.25 = ₹154$ (1).	1+1+1
31	$2x + 3y = 1050$ and $3x + 2y = 1200$ (1). Adding: $5x + 5y = 2250$, so $x + y = 450$; subtracting: $x - y = 150$ (1). So $x = 300$ and $y = 150$ (1).	1+1+1
32	$\angle BDC = \angle BAC = 30^\circ$ (angles in the same segment) (1+1). $\angle CBD = 180^\circ - 100^\circ - 30^\circ = 50^\circ$ (1).	2+1

33	$\pi(14^2 - 7^2) = \frac{22}{7} \times 147 = 462$ cm^2.	2+1
34	$A'(-3,4),\ D'(-7,1),\ M'(-9,6)$. $AD = \sqrt{16+9} = 5$; $A'D' = \sqrt{(-7+3)^2 + (1-4)^2} = \sqrt{16+9} = 5$. Reflection preserves lengths.	1+1+1
35	Area $= (2x+3)^2 = 4x^2 + 12x + 9$ sq. units; perimeter $= 4(2x+3) = 8x + 12$ units.	2+1
	OR: $6 \times 2 = 12$; split $7x = 3x + 4x$: $6x^2 + 3x + 4x + 2 = 3x(2x+1) + 2(2x+1) = (2x+1)(3x+2)$.	1+1+1
36	A: $0.20 \times 7000 = ₹1400$; B: $0.30 \times 4000 = ₹1200$ (2). Family A spends more rupees although B's percentage is higher (1).	2+1
	OR: Except for the first segment, the bars do not start from 0 (1½). So the segment length must be found by comparing end points, which makes estimation harder, though totals are easy to compare (1½).	1½+1½

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) $\frac{5\times200+3\times280}{8} = \frac{1000+840}{8} = ₹230$ per kg (3). (ii) $230 \times 1.2 = ₹276$ per kg (2).	3+2
	OR: (i) $\frac{25+4\times0.5}{5} = \frac{27}{5} = 5.4\%$ (1½). (ii) Yes, 3% lies between 0.5% and 25%: $\frac{25+0.5y}{1+y} = 3 \Rightarrow 25 + 0.5y = 3 + 3y \Rightarrow y = 8.8$ litres (2½). (iii) No; a mixture's concentration always lies between the two concentrations mixed (1).	1½+2½+1
38	(i) $DE = \frac{1}{2}CA = 50$ m, $EF = \frac{1}{2}AB = 30$ m, $FD = \frac{1}{2}BC = 40$ m (1). (ii) Four triangles, all congruent to each other (each is a copy of $\triangle DEF$) (1). (iii) $50 + 30 + 40 = 120$ m (1). (iv) $60^2 + 80^2 = 3600 + 6400 = 10000 = 100^2$, so yes, right-angled at B (2).	1+1+1+2
	OR: (i) $\angle BCD = 180^\circ - 65^\circ = 115^\circ$ and $\angle ADC = 65^\circ$ (1). (ii) $AE = \frac{1}{2}AB = \frac{1}{2}DC = CF$ and $AE \parallel CF$ ⇒ equal parallel sides ⇒ AECF is a parallelogram (2). (iii) $AE = CF = 18$ cm (1). (iv) $2(36 + 20) = 112$ cm (1).	1+2+1+1
39	(i) $1\times19 = 19$, $2\times18 = 36$, $3\times17 = 51$, $4\times16 = 64$, $5\times15 = 75$, $6\times14 = 84$, $7\times13 = 91$, $8\times12 = 96$, $9\times11 = 99$, $10\times10 = 100$ (3). (ii) Greatest 100 (10×10), least 19 (1×19) (1). (iii) It is a square (1).	3+1+1
	OR: (i) If the diameters are $d_1, \ldots, d_n$ then $d_1 + \cdots + d_n = PQ$; length $= \pi(d_1 + \cdots + d_n)/2 = \pi PQ/2$, the same as the big semicircle (3). (ii) $\frac{22}{7} \times \frac{28}{2} = 44$ cm; also $\frac{22}{7} \times \frac{7+7+14}{2} = 44$ cm ✓ (2).	3+2
40	(i) Cubes a^3 and b^3; three cuboids $a \times a \times b$ (vol. a^2b) and three $a \times b \times b$ (vol. ab^2). (ii) $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$. (iii) $5^3 = 125$; $8 + 3(4)(3) + 3(2)(9) + 27 = 8 + 36 + 54 + 27 = 125$ ✓.	2+1+2
	OR: (i) $6x^2 + 11x + 3$. (ii) $96 + 44 + 3 = 143$ m^2 (check $11 \times 13 = 143$ ✓). (iii) $(x+2)^2 = x^2 + 4x + 4$; $(x+3)(x+1) = x^2 + 4x + 3$ — the square is larger by 1 sq. unit.	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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